MILITARY SPECIFICATION

CARRYING CASE, PHOTOGRAPHIC EQUIPMENT FM-22( )

1. SCOPE

1.1 This specification covers one type of photographic carrying case, designated as Carrying Case, Photographic Equipment FM-22( ).

2. APPLICABLE DOCUMENTS

2.1 Documents.- The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

V-T-276	Thread, Cotton
V-T-291	Thread, Linen
NN-P-530	Plywood, Flat Panel
CCC-C-419	Cloth, Cotton, Duck, Unbleached Plied-Yarns (Army and Numbered)
CCC-D-950	Dyeing and Aftertreating Processes for Cotton Fabrics
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple, Wall
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (For Carton Sealing)

FSC 6760

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MIL-P-116	Preservation, Methods of
MIL-C-3432	Cable and Wire, Electrical (Power and Control, Flexible and Extra Flexible, 300 and 600 Volts)
MIL-T-3530	Treatments, Mildew Resistant and Water Repellent For Thread and Twine
MIL-C-3767	Connectors, Electrical (Power, Bladed Type) General Specification For
MIL-S-3950	Switches, Toggle
MIL-A-5433	Adhesive, Application of Room-Temperature and Intermediate-Temperature-Setting Resin (Phenol, Resorcinal, and Melamine Base)
MIL-P-10850	Printer, Projection, Photographic EN-15( )
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-A-46051	Adhesives, Room-Temperature and Intermediate-Temperature Setting Resin (Phenol, Resorcinal, and Melamine Base).

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-169	Extreme Temperature Cycle
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment.

DRAWINGS

ELECTRONICS COMMAND

SC-A-19100	Hardware for Chests and Cases
SC-A-46439	List of Accessories for Package Tester
SC-A-47794	Wood Guide

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

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3. REQUIREMENTS

3.1 Description.- Carrying Case, Photographic Equipment FM-22( ) is a special portable container capable of housing the complete equipment assembly of a photographic set (see 6.6). It is divided into three major compartments. The center section is capable of holding Printer, Projection, Photographic EN-11( ), copying column and lights, and processing trays. The bottom of the case, attached to the center section, is used as an enlarger baseboard and as a base in copy photography. The two adjacent, left-and right-hand, sections are divided into numerous subcompartments and drawers capable of storing all the components of the photographic set in a minimum of space. Installed fixtures include a timer, safelight, two copy lights and column, and a power cable assembly with a control unit.

3.2 Preproduction samples.- The contractor shall furnish two preproduction samples for approval, if required by the invitation for bids and contract. (See 4.3.)

3.3 Baseboard.- The bottom of Carrying Case, Photographic Equipment FM-22( ) shall serve as the base for the enlarger and for copying work. Provisions shall be provided to mount the following equipment on the base board:

(a) Printer, Projection, Photographic EN-11( ) conforming to Specification MIL-P-10850, Type II.

(b) Copying lamps (see 3.4).

(c) Copying column (see 3.5).

3.4 Copying lamps.- Two (2) copy light reflectors on flexible arms approximately one foot long permanently attached to a bracket which can be attached to the baseboard (see 3.3) shall be provided. The diameter of the reflector shall be approximately 6 inches and shall be provided with a switch for turning the lamp on and off at its socket.

3.4.1 Power cable.- A power cable, conforming to Specification MIL-C-3432, shall be provided to provide power to the copying lamps. The cable shall be permanently attached to the copying lamps through the mounting bracket and flexible arms, and the other end shall terminate with Connector, Plug, Electrical Type UP12OM conforming to Specification MIL-C-3767.

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3.5 Copying column.- A removable metal column, approximately 18 inches high, shall be provided for rigid support of camera for copying work. The column shall be of corrosion resisting material or treated to withstand corrosion and shall permit easy vertical adjustment of the copying camera bracket. (See 3.6.)

3.6 Bracket, copying camera.- A lightweight metal camera bracket shall be provided which shall permit the adjustment of the copying camera. The bracket shall extend at least 6 inches higher than the column (see 3.5) and shall permit ready attachment of Camera, Still Picture, Eastman Kodak Company, "Motormatic 35F" or equal, in such a manner as to position the camera parallel to and centered over the base (see 3.3). Provision shall be made to stow the bracket in the case when not in use.

3.7 Cover.- A slip-over type, duck cover shall be provided and shall fasten to the bottom edge of the case, in the closed position, by means of snap fasteners.

3.7.1 Duck.- Duck shall conform to Specification CCC-C-419, Type III, and shall be given class B after treatment, using inhibitors "a", "e," "f," or "g," as specified in Specification CCC-D-950; inhibitor "a" shall be used when duck will come in contact with natural or synthetic rubber.

3.7.2 Thread.- Cotton and linen thread shall conform to Specification V-T-276 and V-T-291, respectively, and shall be given mildew resistant treatment in accordance with Specification MIL-T-3530.

3.7.3 Color.- The color of fabric shall be Olive-Drab No. 7. The color of thread shall match shade S-1 of the United States Army Standard Color Card for Official Standardization Shades of Sewing Thread. (See 6.5.)

3.7.4 Shrinkage.- Fabric and thread shall be pre-shrunk, or allowance shall be made for shrinkage, in order to provide for satisfactory fit of finished items before and after they are immersed in water and then dried. (See 4.8.)

3.8 Control unit and power cable.- A control unit, consisting of three toggle switches conforming to Specification MIL-S-3950, shall be provided. The switches shall control the following: (1) enlarger, (2) accessory receptacle, and (3) safelight. The power cable, conforming to Specification MIL-C-3432, shall be 10 feet in length, and shall be permanently attached to the control unit. The free end shall terminate with Connector, Plug, Electrical Type UP120M conforming to Specification MIL-C-3767.

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3.9 Safelight and timer.- A safelight and a mechanical timer shall be installed in the case.

3.10 Storage compartments.- Storage compartments shall be provided to house all the components of photographic set. (See 6.6.)

3.11 Hardware.- Hardware for the case shall conform to Drawing SC-A-19100.

3.12 Plywood.- Plywood shall be Douglas fir or hardwood plywood conforming to Specification NN-P-530 and Drawing SC-A-47794.

3.13 Adhesives.- Wood-to-wood joints shall be made with glue conforming to Specification MIL-A-46051 and the glue shall be applied in accordance with Specification MIL-A-5433.

3.14 Dimensions.- Carrying Case, Photographic Equipment FM-22( ), when closed, shall not exceed 23 inches high by 17 inches wide by 24 inches deep.

3.15 Weight.- The weight of the carrying case shall not exceed 60 pounds.

3.16 Finish, protective.- Equipment shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc, and necessary touch up after mounting. The final paint film on type I surfaces shall be green color (olive drab), semigloss enamel matching a color chip provided by the procuring agency. (See 6.3.)

3.17 Marking.- Marking shall conform to Specification MIL-M-13231.

3.18 Service conditions.- The equipment shall meet the following service conditions--where a test is referenced, meeting the test shall be considered as compliance with the requirement.

3.18.1 Temperature (see 4.9).- Exposure in the range of +160°F to -80°F.

3.18.2 Relative humidity (see 4.7).- Exposure up to 97 percent relative humidity for 20 hours and 100 percent relative humidity, with condensation, for 4 hours for each 24-hour period.

3.18.3 Bounce.- After subjection to the test specified in 4.6, there shall be no physical damage to the equipment and no component shall be loose.

3.19 Workmanship.- The equipment shall be manufactured and assembled in accordance with the requirements specified herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3.)

(b) Inspection covered by subsidiary documents. (See 4.4.)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery. (See 4.11.)

4.3 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A and group B. (See paragraphs 4.3.1 and 4.3.2 respectively.)

Table I.- Preproduction inspection

Inspection (For additional preproduction inspection see 4.3)	Req Para	Insp Para
Bounce	3.18.3	4.6
Moisture resistance	3.18.2	4.7
Temperature	3.18.1	4.9

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4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Finish	3.16
Marking	3.17
Glue application	3.13

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.3. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.7.) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection.- This inspection shall consist of the inspection specified for Visual and mechanical in paragraph 4.10 and shall conform to the inspection procedures of Standard MIL-STD-105 using the general inspection levels. The Acceptable Quality Level (AQL) shall be 1.0 percent for major defects and 4.0 percent for minor defects.

4.5.2 Group B inspection.- This inspection shall consist of the inspection specified for Shrinkage in paragraph 4.8 and shall conform to the procedures for small-sample inspection of Standard MIL-STD-105 using the special inspection levels. The AQL shall be 4.0 percent defective and the inspection level shall be S-4 for normal, tightened and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.5.3 Reinspection of conforming group B sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B inspection may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.6 Bounce test.- The equipment shall be tested on the package tester, type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. Accessories shall be selected from those listed on Drawing SC-A-46439. The test shall be as follows:

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(a) Cover the tester bed with a panel of 1/2 inch plywood, with the grain parallel to the drive chain. Space sixpenny nails with the heads below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

(b) Place the case, containing all the components of the photographic set on the bed of the package tester. Limit the lateral motion, by wood fences, to not more than 3 inches and not less than 1 inch. Additional barriers may be used to prevent tumbling, provided that the fore-and-aft motion of the equipment against the backstop is not restrained.

(c) Operate the package tester, shafts in phase, for a total of 3 hours at 284 ±2 rpm. Turn the equipment at the end of each 30 minutes so it will rest on a new face.

4.7 Moisture resistance test.- The equipment shall be tested as follows:

(a) Dry at 130° ±5°F for 24 hours.

(b) Condition at 77° ±5°F and 40 to 50 percent relative humidity for 24 hours.

(c) Inspect the equipment for conformance with specified requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity and period of time for each portion of the cycle shall conform to Standard MIL-STD-170.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ±5°F and 40 to 60 percent relative humidity. No repair or replacement of parts shall be made.

(f) If the equipment fails to meet full specification requirements, as specified in (e) above, it does not pass the test.

4.8 Shrinkage test.- The cover shall be tested to determine compliance with the requirement specified in 3.7.4.

4.9 Temperature test.- The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169. After cycling is completed, the equipment shall be inspected for any defects.

4.10 Visual and mechanical inspection.- Equipment shall be examined for the defects listed in table II.

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Table II.- Classification of visual and mechanical defects

Classification	Defect
Major	Component missing Fastening device missing Cover too tight Switch inoperative
Minor	Dimensions and weight not as specified Color not as specified Scratches in finish

4.11 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine compliance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be Level A or Level C as specified. (See 6.2.)

5.1.1 Level A.- Carrying Case, Photographic Equipment FM-22( ) shall be packaged in accordance with the requirements of Specification MIL-P-116, for the method as specified as follows:

5.1.1.1 Prepackaging.- Carrying Case, Photographic Equipment FM-22( ) shall be pre-packaged as follows: Copy lights, bracket, and copy column shall be cushioned with single faced flexible corrugated fiberboard, conforming to Specification PPP-P-291. Secure cushioning with gummed paper tape. Cushion timer and safe-light by placing a pod of cellulose wadding or flexible corrugated paper over the timer and safelight to prevent glass breakage. Stow the cushioned items within the case. Cushion all other items in the case to prevent movement. Close the case and secure fastenings.

5.1.1.2 Carrying Case, Photographic Equipment FM-22( ) shall be packaged Method IC-5, as follows: Cushion the case on all surfaces with cells and pads of fiberboard designed to absorb the shock or impact normally encountered in handling and transit. Place the cushioned case within a close-fitting fiberboard box, Class Weather resistant W5c or W6c. Seal the box closure with water resistant pressure-sensitive tape, conforming to Specification PPP-T-76.

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5.1.2 Level C.- Carrying Case, Photographic Equipment FM-22( ) shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing.- Packing shall be level A, level B, or level C as specified. (See 6.2.)

5.2.1 Level A.- Carrying Case, Photographic Equipment FM-22( ), packaged as specified above, shall be packed within nailed wood, wood-cleated plywood, or triple wall fiberboard boxes conforming to Specifications PPP-B-621, PPP-B-601, and PPP-B-640, respectively. Gross weight of shipping containers shall not exceed 200 pounds. Shipping containers shall be closed and strapped as specified in the applicable box specification.

5.2.2 Level B.- Carrying Case, Photographic Equipment FM-22( ), packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the shipping containers shall be domestic type and no strapping required.

5.2.3 Level C.- Carrying Case, Photographic Equipment FM-22( ), packaged as specified in 5.1, shall be packed in shipping containers in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

5.3 Marking.- Unit packages and shipping containers shall be marked in accordance with requirements of Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- Carrying Case, Photographic Equipment FM-22( ) is used to house the components of a photographic set capable of producing 35 millimeter negatives and projection prints.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

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(b) Level of packaging and level of packing required for shipment (Level A, level B, or level C).

(c) Marking and shipping of samples.

(d) Place of final inspection.

6.3 Color.- The color ship furnished by the procuring agency (see 3.16) will be color chip No. X-24087 and may be obtained upon request to: Commanding General, U. S. Army Electronics Command, ATTN: AMSEL-PP-CP, 225 South Eighteenth Street, Philadelphia, Pennsylvania 19103.

6.4 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: FM-22W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.5 Color of textiles.- Standard samples of Olive Drab No. 7 may be obtained from the Sample Loan Unit, Military Clothing and Textile Supply Agency, Philadelphia, Pennsylvania 19103. When requesting samples, indication should be given as to type of material concerned.

6.6 Phogotraphic set.- The photographic set to be housed in the carrying case shall consist of the following components:

1 ea	Printer, Projection, Photographic EN-11( )
1 ea	Camera, Still Picture (35 millimeter)
1 ea	Holder FN-9( )
6 ea	Clip, Photographic Film
1 ea	Meter, Photographic Exposure
1 ea	16 oz graduate
1 ea	Cable Release, Still Camera
2 ea	Print Paddle
3 ea	Tray, print size 8 x 10
1 ea	Thermometer
1 ea	Tank, PH-322
1 ea	Sponge, cellulose
6 packages	Paper, lens tissue
25 yards	Cheesecloth
1 ea	Ruler, 12 inch
1 ea	Filter, light, lightyellow
1 ea	Ring, retaining
2 ea	Lens, for close-up
1 ea	Clamp, for mounting camera
1 ea	Hood, lens

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1 ea	Flasher, flashlamp with 3-inch and 4-inch reflectors
1 ea	Instruction book
2 ea	Technical manual
18 ea	Film, 35 mm
6 ea	Battery (flash)
24 packs	Developer
6 packs	Hardener and fixer
6 packs	Enlarging paper
30 ea	Blotter
16 ea	Lamp, (photoflash)
9 ea	Lamp, (enlarger)
3 ea	Lamp (safe light)

6.7 Verification isnpection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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